Amit

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Smart Contract & Web3 Engineer specializing in DeFi protocols, account abstraction, and cross-chain systems. Independently designed and deployed production contracts across perpetuals DEXs, DAO governance, ERC-4337, soulbound identity, and stablecoins. Every system is invariant-tested, verified on-chain, and shipped with a production-grade frontend.

TECHNICAL SKILLS

Contracts
Solidity 0.8, Foundry, OpenZeppelin, EVM internals, gas optimization, storage slot packing

Protocols
ERC-20/721/4337/5484, EIP-712, EIP-1967 UUPS, Chainlink VRF / CCIP / Price Feeds, Aave V3

Testing
Unit, Integration, Stateless Fuzz, Stateful Invariant Fuzzing, CEI pattern, flash-loan resistance

Frontend
Next.js 14/15, TypeScript, Wagmi v2, Viem, Ethers.js, RainbowKit, TanStack Query v5

Infra
Vercel, Node.js, IPFS / Pinata, Git, Foundry DevOps scripts, multi-chain deployment

PROJECTS & ENGINEERING WORK

Jan 2026

Nexus Perpetuals DEX — Ethereum Sepolia

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- On-chain perp exchange — 50× leverage, `PerpsVault.sol` dual accounting, ERC-4337 `NexusPaymaster` for 100% gasless trading, Chainlink CCIP cross-chain margin with nonce deduplication
- 95 tests + 6,400 invariant mutations — zero solvency violations; Next.js 14 + Wagmi v2 frontend, zero backend

Jan 2026

Nexus Perpetuals DEX v2 — Polkadot Hub

Smart Contract + Full-Stack Developer · Solo · *Polkadot Solidity Hackathon 2026 — Participant*

- Full Nexus architecture ported to Polkadot Hub Testnet — live Binance WebSocket price sync, CCIP cross-chain margin, open-keeper liquidation; same 6,400-mutation invariant suite

Late 2025

Sentinel DAO

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- Modular governance OS — 48H `DAOTimelock`, `VetoCouncil` + `RageQuit` minority exit, `TreasuryYieldStrategy` auto-compounding into Aave V3, ERC-4337 `SessionKeyModule` for gasless voting
- 256 tests, zero failures across unit, integration, and fuzz

Early 2026

RST Protocol — ERC-5484 Soulbound Reputation

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- On-chain reputation via ERC-5484 SBT — dynamic SVG medals (5 tiers), zero IPFS dependency, UUPS upgradeable engine + immutable token, score clamped [0, 1000] via pure `ReputationMath` library
- `ReputationVault` with 12h/24h cooldowns, strict CEI, `nonReentrant`; fuzz tested — overflow and underflow impossible by design

Mid 2025

ERC-4337 Account Abstraction & UUPS Protocol

Smart Contract Developer · Solo

- ERC-4337 from raw spec — `validateUserOp`, custom `SimplePaymaster`, session key delegation; EIP-1967 proxy with atomic V1→V2→V3 migration using `__gap` arrays

Early 2025

Decentralized StableCoin & Merkle-712 Airdrop

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- 200% overcollateralized stablecoin (wETH/wBTC) with Chainlink oracles; solvency proved across 10,000+ fuzz sequences
- O(1) Merkle airdrop for 1M+ recipients — EIP-712 sigs bind to `msg.sender` + `chainId`, phased vesting with burn-on-expiry

2024

Core Protocol Implementations

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- Provably Fair Raffle (Chainlink VRF + Automation) · SISO ERC-20 (RBAC, Pausable, Sepolia verified) · CuteCat NFT (ERC-721, IPFS) · Oracle Crowdfunding (Chainlink feeds, 1e18 precision)

EDUCATION & CERTIFICATIONS

2024

Cyfrin Updraft — Smart Contract & Security Engineering

Certification · Completed in Full · Self-Directed

- EVM internals, storage slot packing, gas optimization, Foundry invariant fuzzing, CEI + flash-loan resistance — applied from line one across every project
- Ongoing self-study — ERC-4337, ERC-5484, UUPS, DAO systems, AMM math, Chainlink CCIP from raw EIP specs; every concept shipped as verifiable production code